



sAifer Lab
Joint lab on Safety and Security of AI

Deepfakes, Face Morphing and Presentation Attack: Generalization Challenges and Robust Detection in the Wild

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13 Maggio 2026

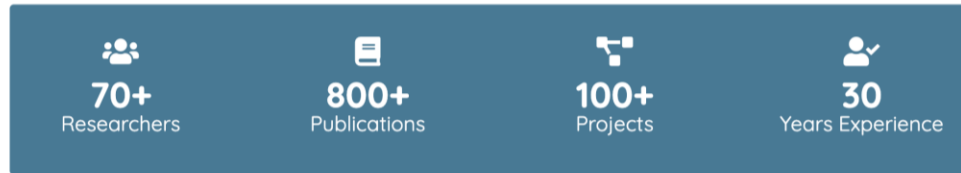


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University of Cagliari – PRA Lab & Biometric Unit

- <https://www.saiferlab.ai/research/biometrics>



2 Joint Research Laboratories



PRA LAB
Department of Electrical
and Electronic Engineering (DIEE)
University of Cagliari



SMARTLAB
Department of Informatics Bioengineering, Robotics,
and System Engineering (DIBRIS)
University of Genoa



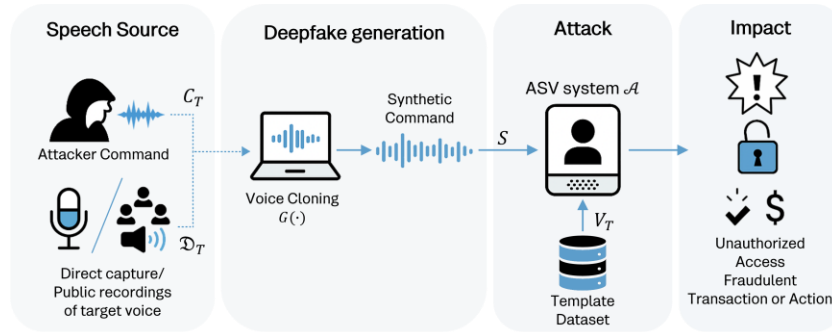
Outline of the talk

- Security in biometric systems
- Generalization issues: three case studies
- sAlfer Lab experience
- Conclusions



Biometric Security: Presentation Attacks, Morph & Deepfake Attacks

- Presentation attacks: Present artificial replicas to a sensor
- Deepfake attack: piece of synthetic media such as video, audio recording, or image, that has been digitally manipulated



Biometric Security: Presentation Attacks, Morph & Deepfake Attacks

Subject A
(Bonafide)



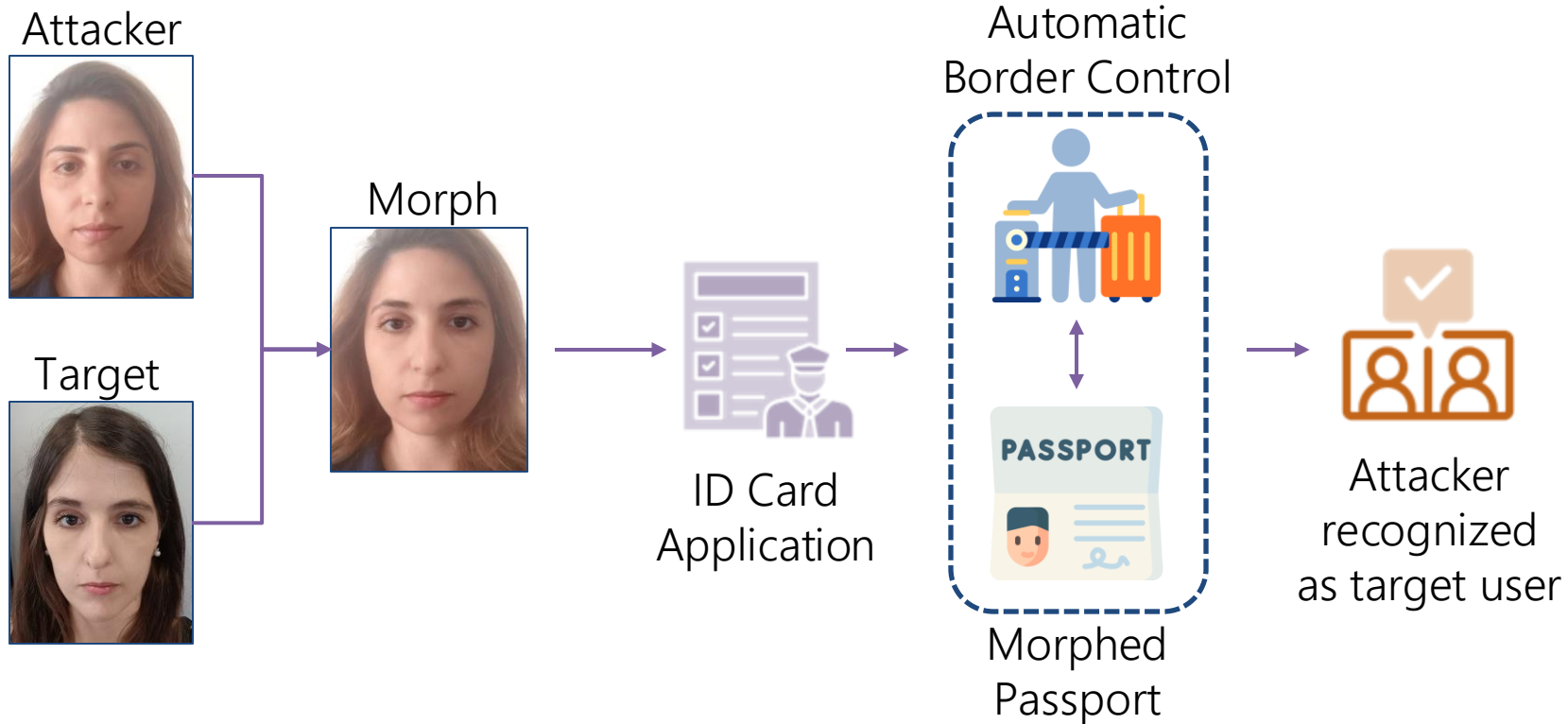
Subject B
(Impostor)

Morphed image (blending 50%)

"A morph is a seamless transformation of an image into an other"

Morph as attack: Intermediate images contains the biometric characteristics of both individuals a can be used to carry out presentation attacks on face recognition systems

Biometric Security: Morph Attack Scenario



M. Ferrara, A. Franco and D. Maltoni, "The magic passport," IEEE International Joint Conference on Biometrics, Clearwater, FL, USA, 2014, pp. 1-7, doi: 10.1109/BTAS.2014.6996240.



«Passive» Detection Problem

- Any methodology (black or white box) able to detect attacks (deepfake, morph or presentation attack) in a video or in an image
- The source is supposed to come without any previous malicious actions except from the deep learning-based digital manipulations

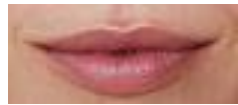


«Passive» detector

- A posteriori probability
- Most likely regions/frames

Presentation Attack Detection (PAD) / Deepfake Detection

Presentation Attack Detector/ Deepfake Detector



Input

**Biometric
sensor**

**Feature
extraction**

Feature set

Classifier

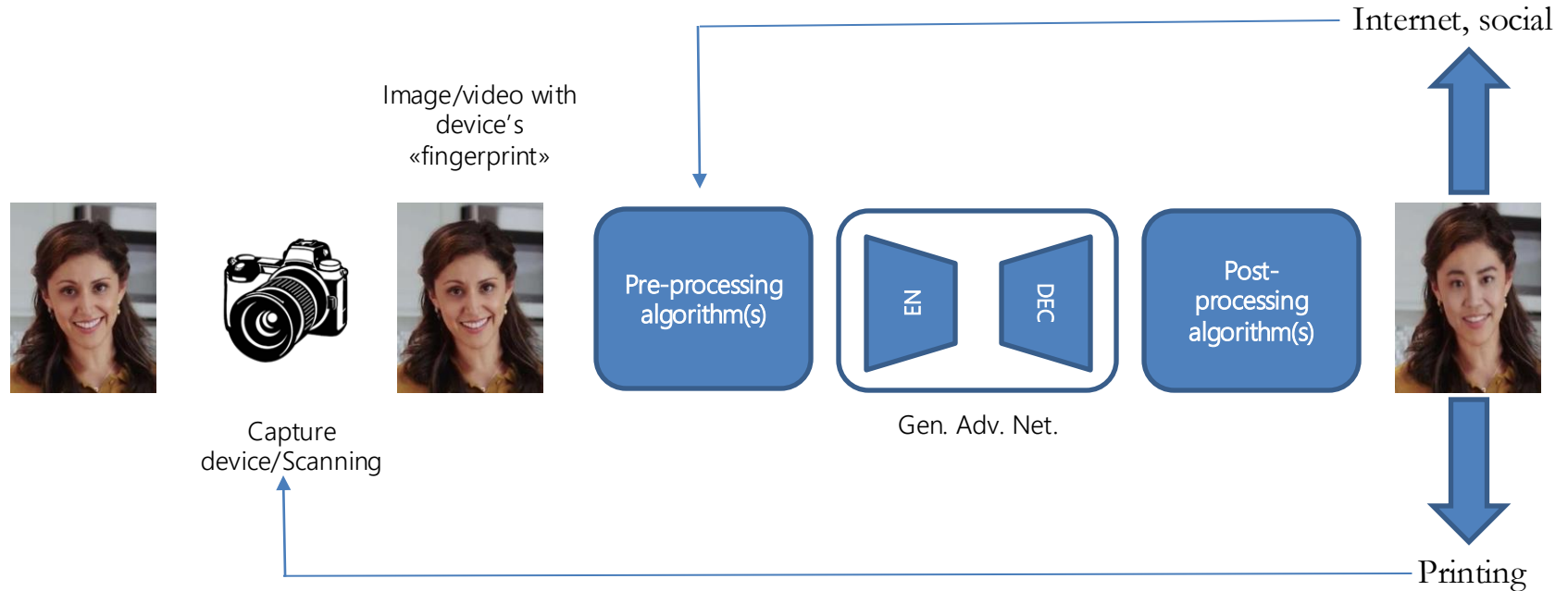
Liveness score



Database

Presentation Attack Detection (PAD) / Deepfake Detection

A very difficult task



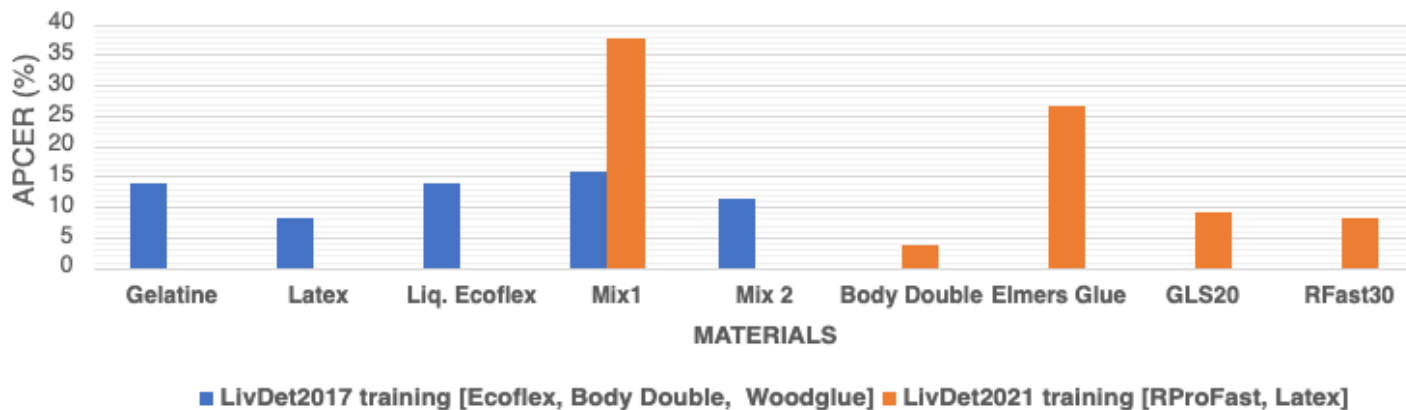
P. Korshunov and S. Marcel, Improving Generalization of Deepfake Detection With Data Farming and Few-Shot Learning, *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 4 (3) 386-397 2022, doi: 10.1109/TBIOM.2022.3143404.

Generalization issue: Case Study 1 – Fingerprint PAD



Why do PAD models fail?

- Models struggle with never-seen-before attacks
- Poor generalization beyond training data

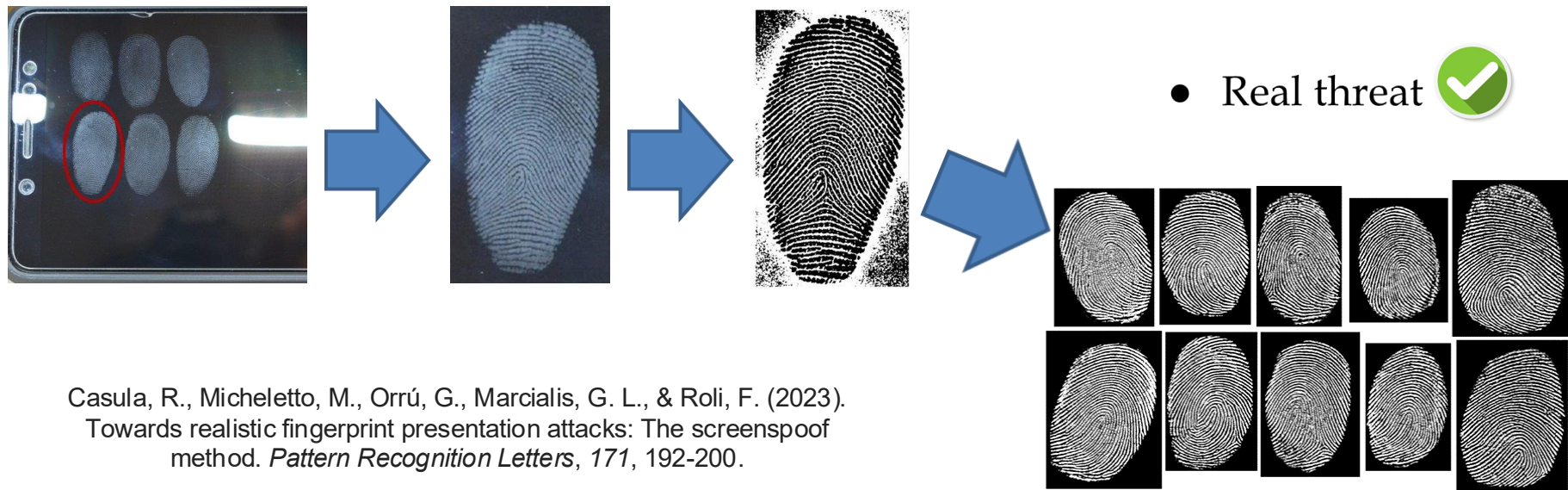


Micheletto, M., Orrù, G., Casula, R., Yambay, D., Marcialis, G. L., & Schuckers, S. (2023). Review of the fingerprint liveness detection (livdet) competition series: from 2009 to 2021. *Handbook of biometric anti-spoofing: presentation attack detection and vulnerability assessment*, 57-76.

Generalization to never-seen-before attacks remains difficult!

Generalization issue: Case Study 1- Why PAD models fail?

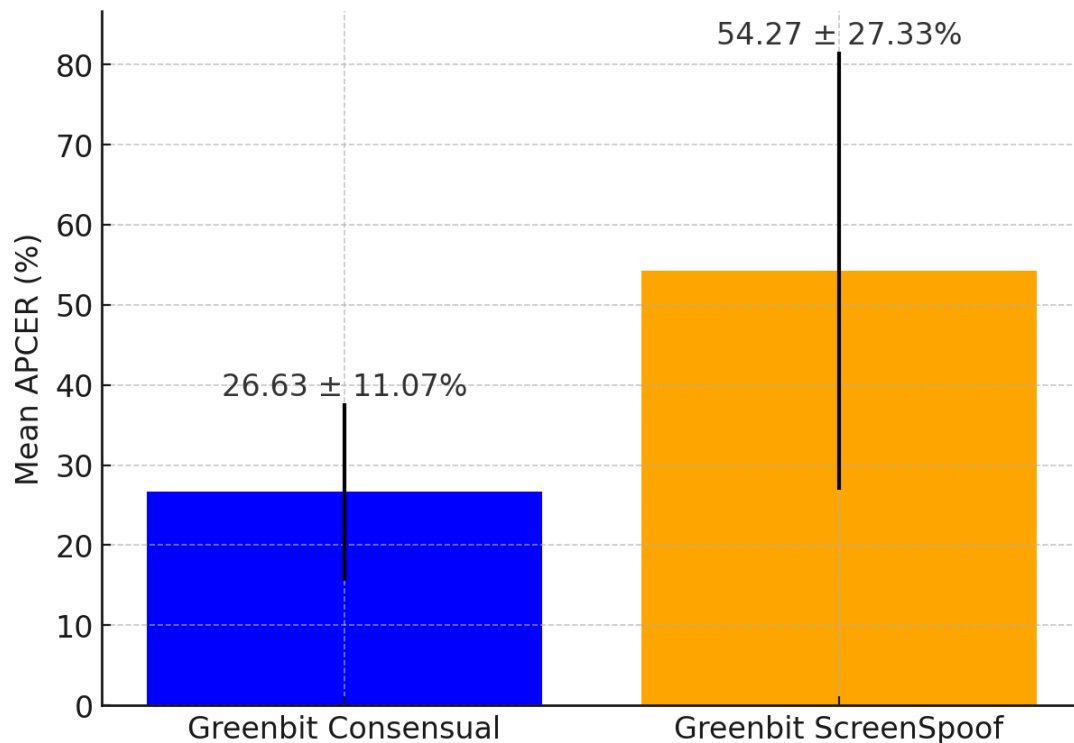
A specific example: the ScreenSpoof case



Casula, R., Micheletto, M., Orrú, G., Marcialis, G. L., & Roli, F. (2023). Towards realistic fingerprint presentation attacks: The screenspooft method. *Pattern Recognition Letters*, 171, 192-200.

Generalization issue: Case Study 1 - Why PAD models fail?

PADs trained on the basic molding and casting method struggle to generalize across different replica creation techniques



Generalization issue: Case Study 2 – Image Deepfake detection

Why do deepfake models fail?

- Different deepfake generation models
- Differences in image size, compression, quality, etc. across different datasets

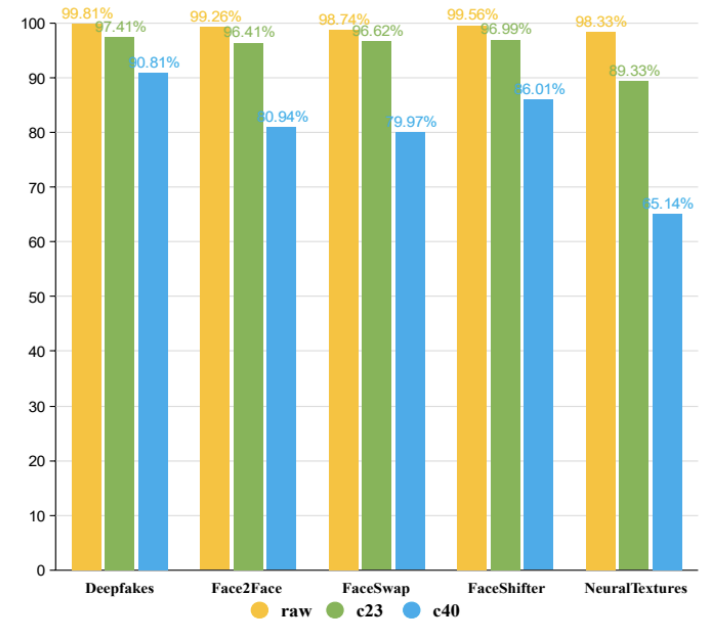
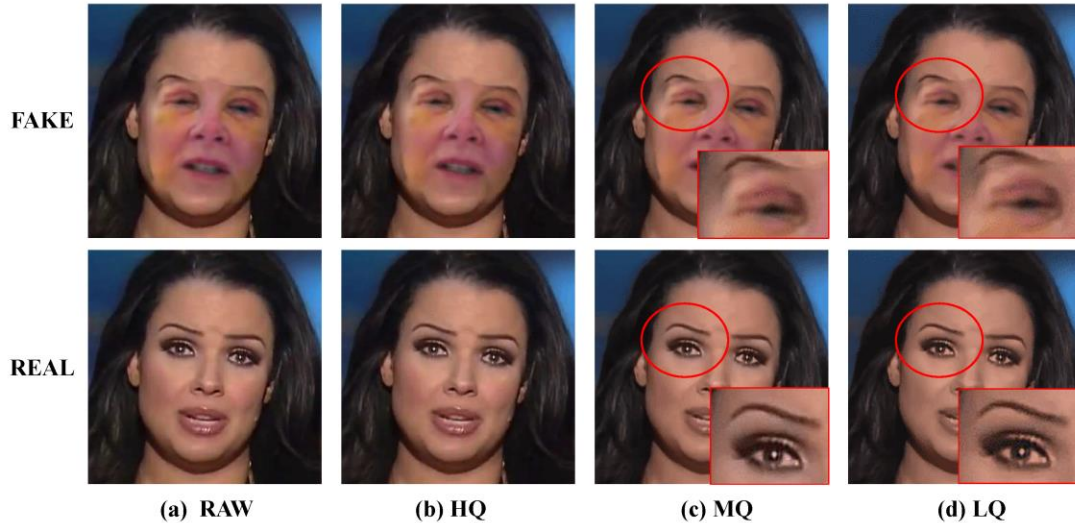
frame-level results	Test set										Avg.
Method	Intra-dataset (FF++)						Cross-dataset				
	DF	F2F	FS	FSH	NT	Intra Avg.	WildDF	CelebDF	DFDC	Cross Avg.	
Xception	83.73	81.87	79.91	77.99	79.08	80.52	58.03	66.35	57.23	60.54	73.02
DSP-FWA	70.48	66.86	60.21	69.14	60.15	65.37	53.39	57.11	55.02	55.17	61.54
B4	91.76	89.62	88.35	88.09	87.14	88.99	63.47	73.75	59.30	65.51	80.18
B4ATT	92.63	91.00	87.78	89.36	86.81	89.52	62.65	69.29	58.28	63.41	79.72
MCX-API	71.44	67.46	64.95	68.66	63.43	67.19	50.78	56.43	54.94	54.05	62.26
Our (E-TAD)	97.11	96.04	91.58	95.24	86.61	93.32	64.12	64.42	56.40	61.65	81.44

Gao, Jie, et al. "Texture and artifact decomposition for improving generalization in deep-learning-based deepfake detection." *Engineering Applications of Artificial Intelligence* 133 (2024): 108450.

Generalization to never-seen-before attacks remains difficult!

Generalization issue: Case Study 2 – Image Deepfake detection

FaceForensics++ dataset: different compression levels: RAW, HQ (c23), MQ() and LQ (c40)

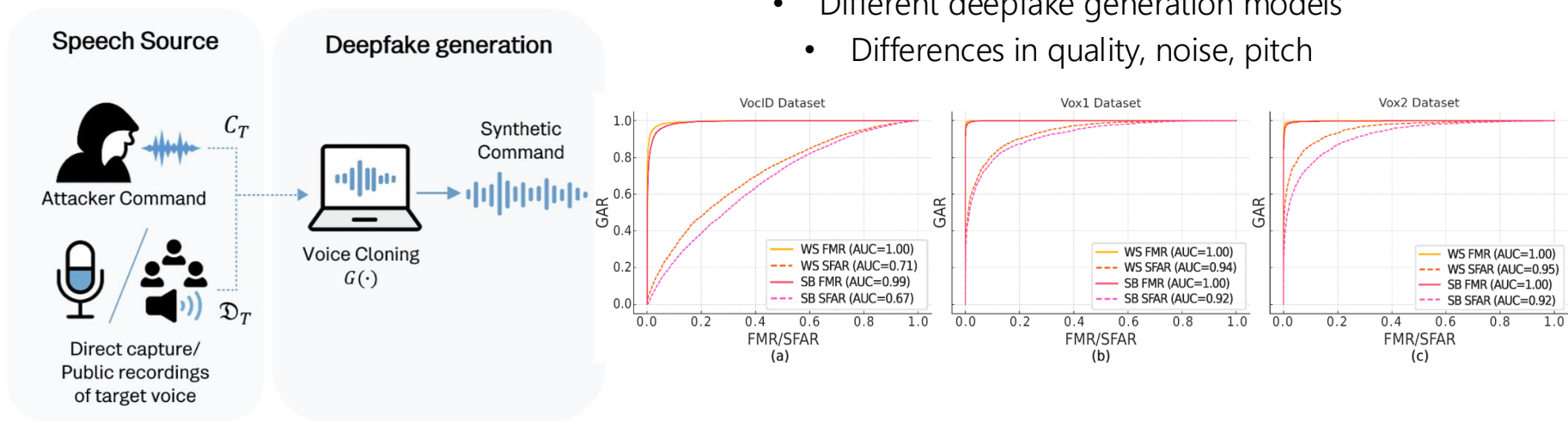


Gao, Jie, et al. "DeepFake detection based on high-frequency enhancement network for highly compressed content." *Expert Systems with Applications* 249 (2024): 123732.

Generalization issue: Case Study 3 – Audio Deepfake detection

Why deepfake models fail?

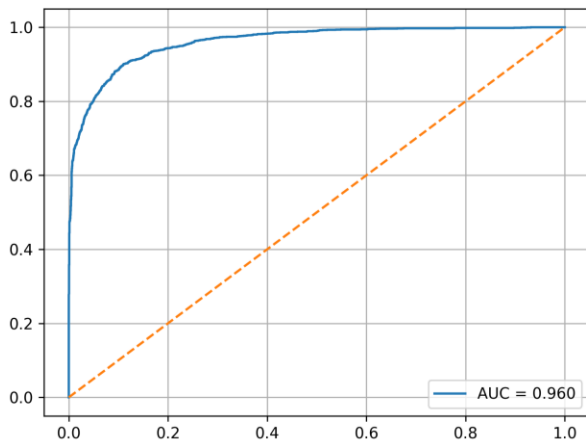
- Different deepfake generation models
- Differences in quality, noise, pitch



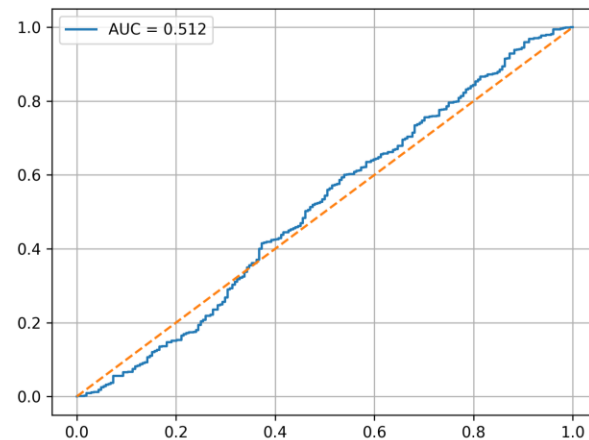
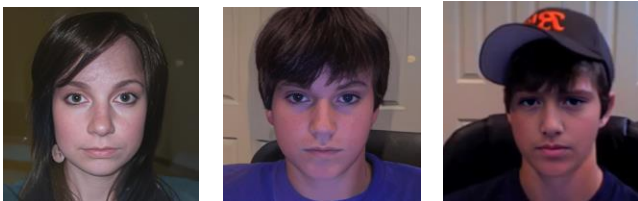
Micheletto, M., et al. "Deepfake Attacks on Automatic Speaker Recognition Systems Using a Novel Dataset of Realistic Voice Commands", under review.

Generalization to never-seen-before attacks remains difficult!

Generalization issue: Case Study 4 – Morph Attack detection



Intra-dataset



Cross-dataset

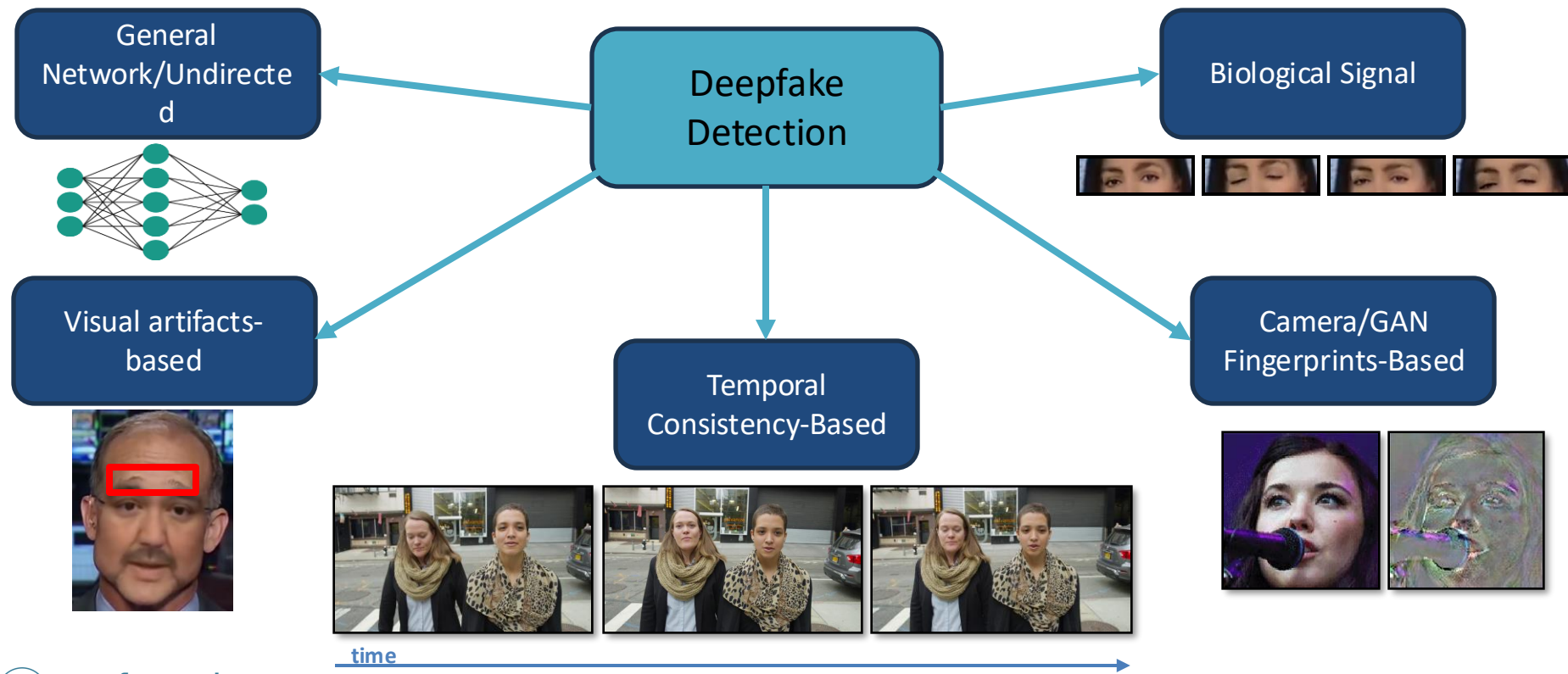


Generalization to never-seen-before attacks remains difficult!

The sAlfer Lab experience on DeepFake deteccion



Deepfake and Morph Detection Taxonomy

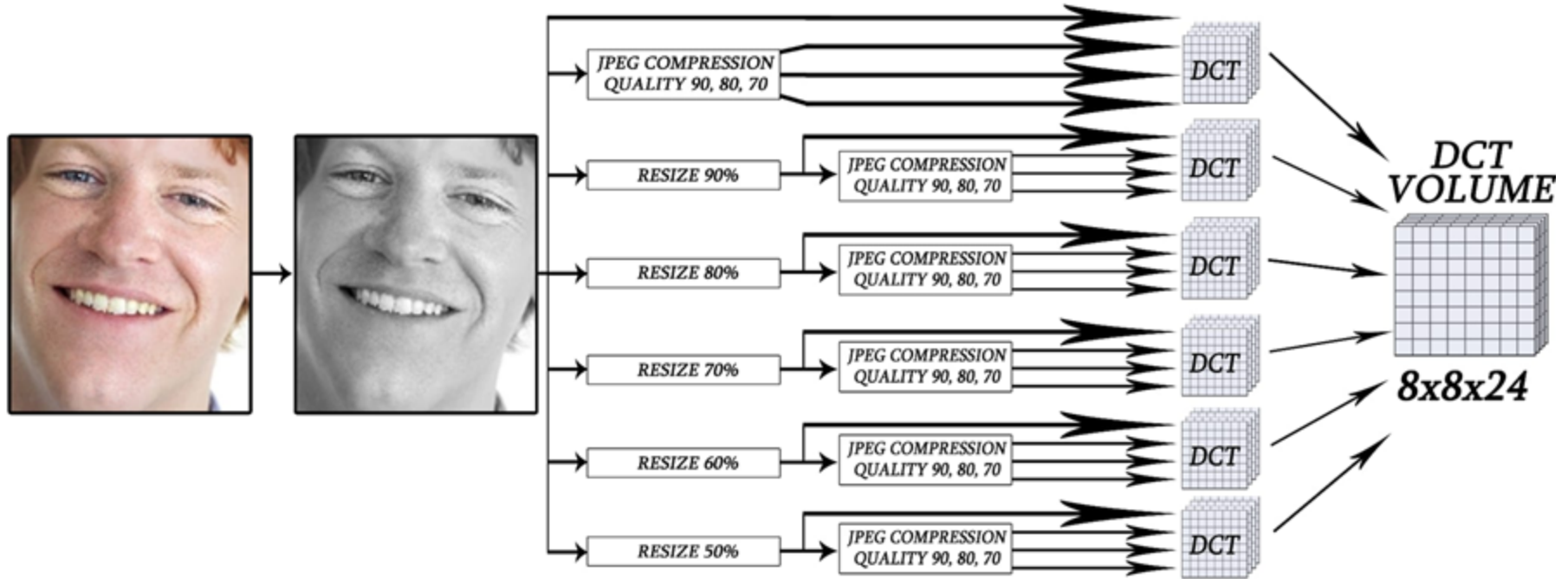


Deepfakes in the real life

- **Realistic scenarios often include some additional transformations applied to the input image**
 - Compression
 - Resizing
- **Lack of generalization**



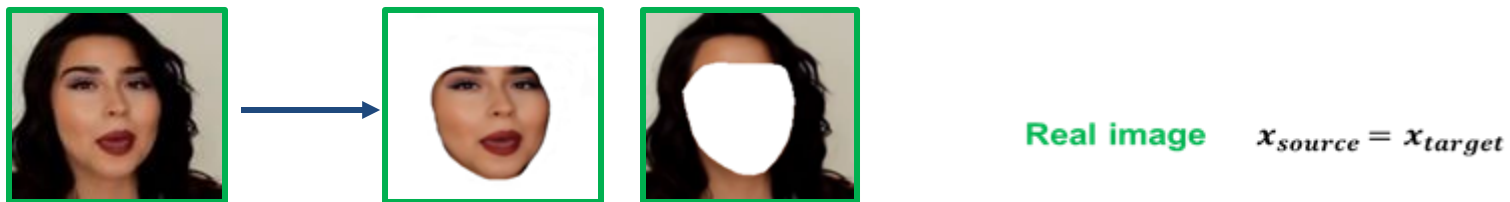
Embedding scale and resize operations: the proposed tensor-based representation



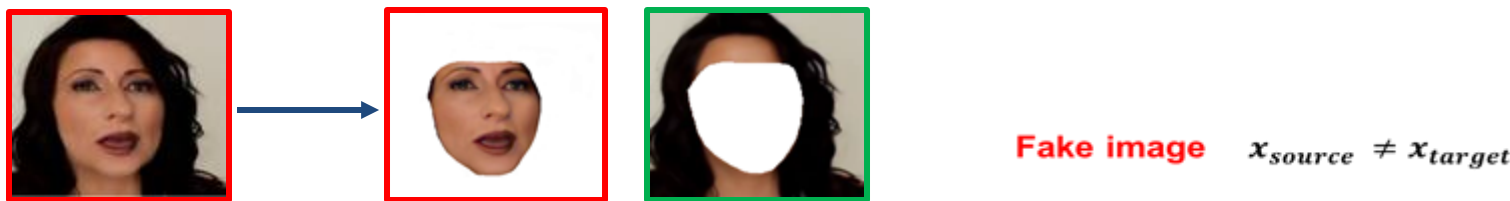
[S. Concas, G. Perelli, G. L. Marcialis and G. Puglisi, "Tensor-Based Deepfake Detection in Scaled and Compressed Images," 2022 IEEE International Conference on Image Processing (ICIP), Bordeaux, France, 2022]

Generalized Deepfake Detection Algorithm Based on Inconsistency Between Inner and Outer Faces

Assumption: a fake image is composed of two parts, one related to the source image and the other related to the target image



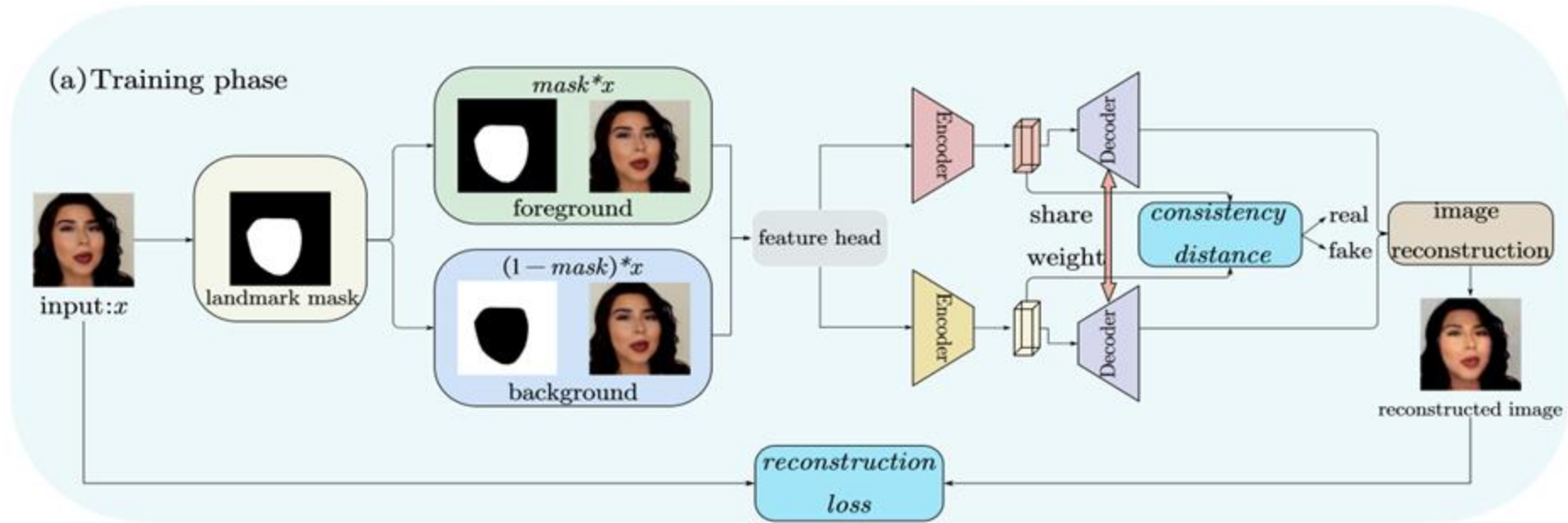
Inner Face (Target) Outer Face (Source)



Inner Face (Target) Outer Face (Source)

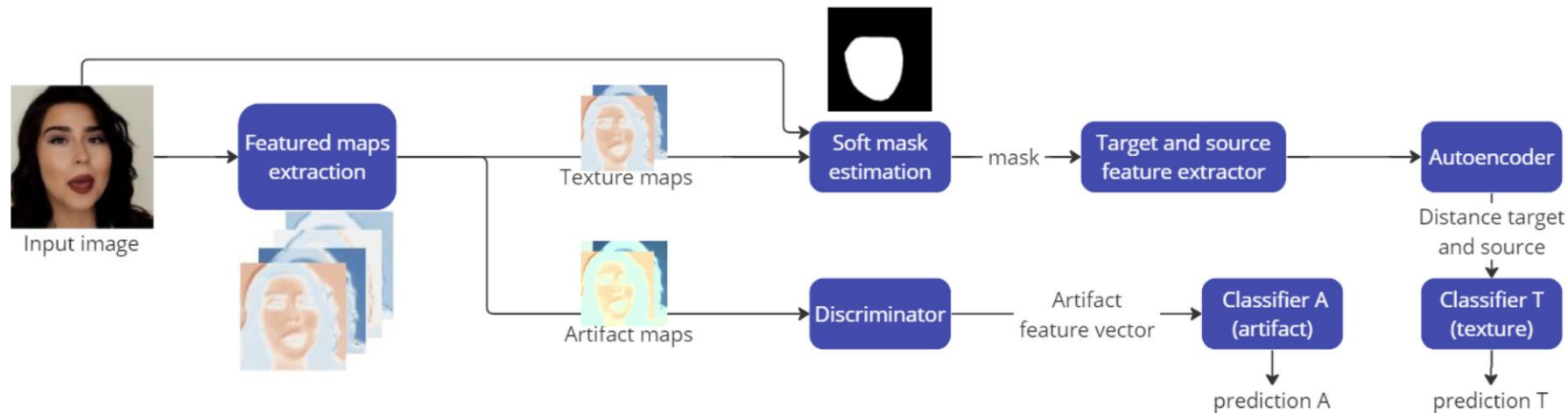
[Gao, J., Concas, S., Orrù, G., Feng, X., Marcialis, G.L., Roli, F. (2024). Generalized Deepfake Detection Algorithm Based on Inconsistency Between Inner and Outer Faces. In: Foresti, G.L., Fusiello, A., Hancock, E. (eds) Image Analysis and Processing - ICIAP 2023]

Generalized Deepfake Detection Algorithm Based on Inconsistency Between Inner and Outer Faces



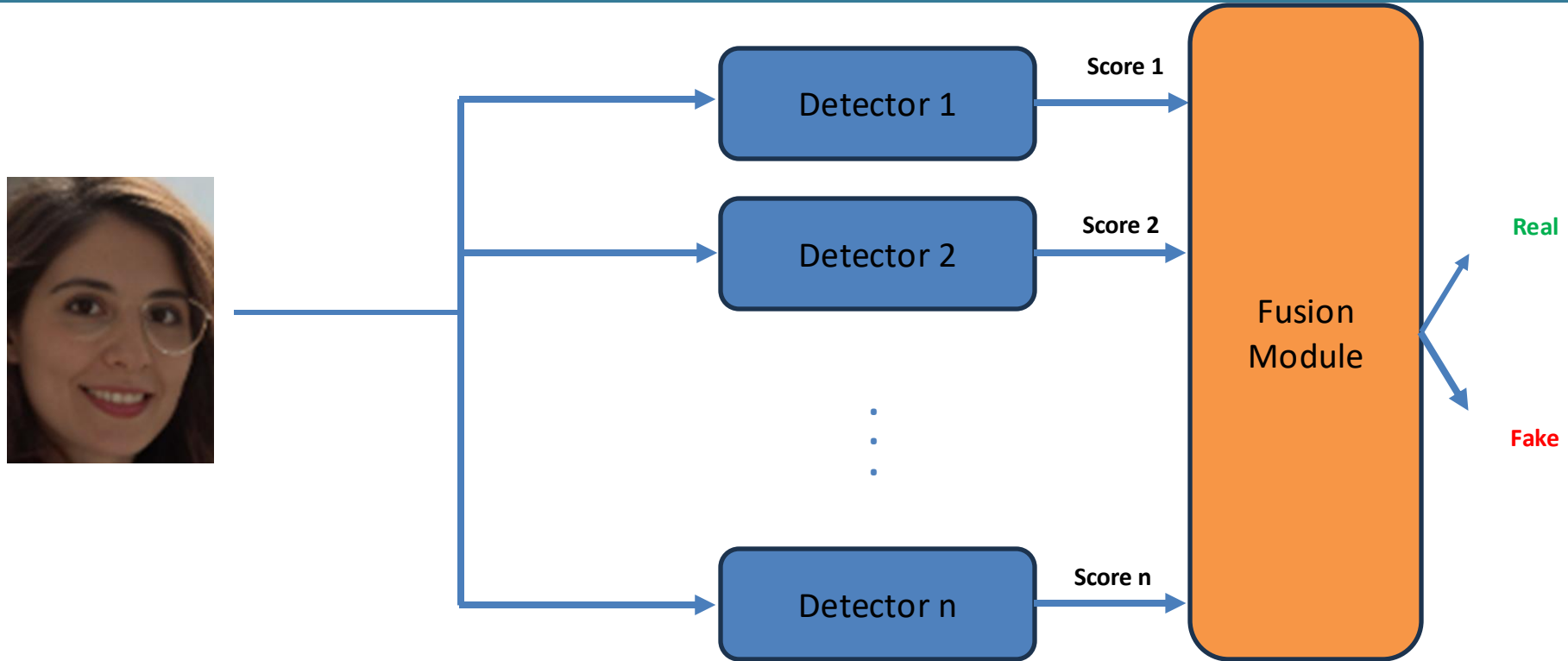
[Gao, J., Concas, S., Orrù, G., Feng, X., Marcialis, G.L., Roli, F. (2024). Generalized Deepfake Detection Algorithm Based on Inconsistency Between Inner and Outer Faces. In: Foresti, G.L., Fusiello, A., Hancock, E. (eds) Image Analysis and Processing - ICIAP 2023]

Texture and artifact decomposition for improving generalization in deep-learning-based deepfake detection



[Jie Gao, Marco Micheletto, Giulia Orrù, Sara Concas, Xi-aoyi Feng, Gian Luca Marcialis, and Fabio Roli. Texture and artifact decomposition for improving generalization in deep-learning-based deepfake detection. Engineering Applications of Artificial Intelligence]

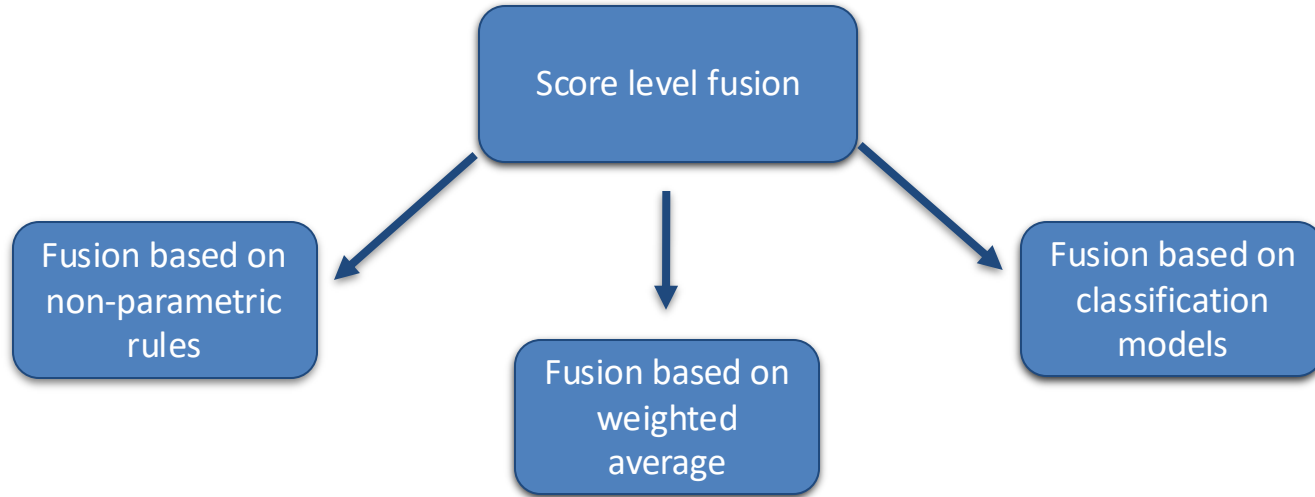
Multi-modal based deepfake detection



[Concas, S.; La Cava, S.M.; Orrù, G.; Cuccu, C.; Gao, J.; Feng, X.; Marcialis, G.L.; Roli, F. Analysis of Score-Level Fusion Rules for Deepfake Detection. Appl. Sci. 2022]

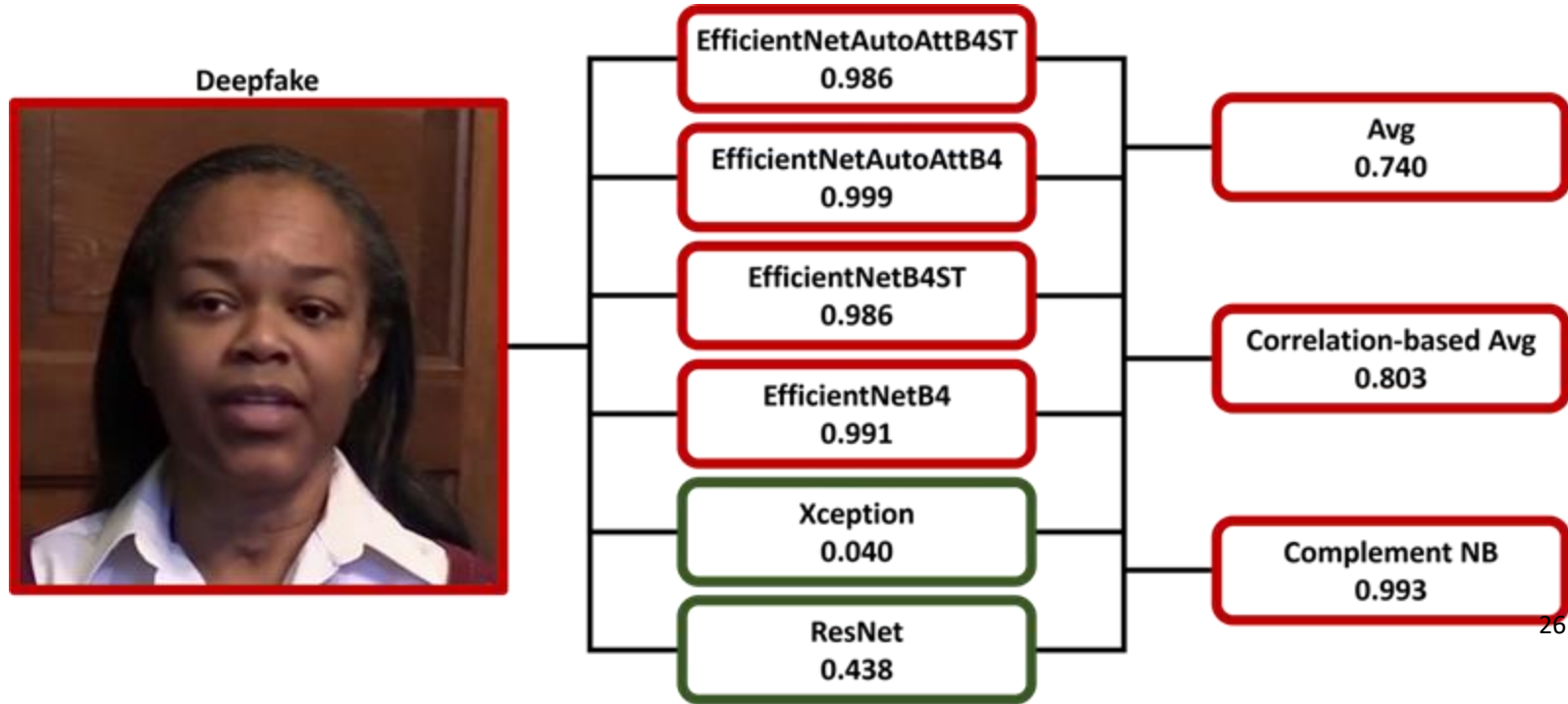
Multi-modal based deepfake detection

Analyzed Score-level fusion methods



[Concas, S.; La Cava, S.M.; Orrù, G.; Cuccu, C.; Gao, J.; Feng, X.; Marcialis, G.L.; Roli, F. Analysis of Score-Level Fusion Rules for Deepfake Detection. Appl. Sci. 2022]

Multi-modal based deepfake detection: Classification Example



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[Concas, S.; La Cava, S.M.; Orrù, G.; Cuccu, C.; Gao, J.; Feng, X.; Marcialis, G.L.; Roli, F. Analysis of Score-Level Fusion Rules for Deepfake Detection. Appl. Sci. 2022]

Quality-based Artifact Modeling for Facial Deepfake Detection in Videos

Problem modelling



Some deepfakes present visible artifacts

Hypothesis: artifacts are present even when not visible

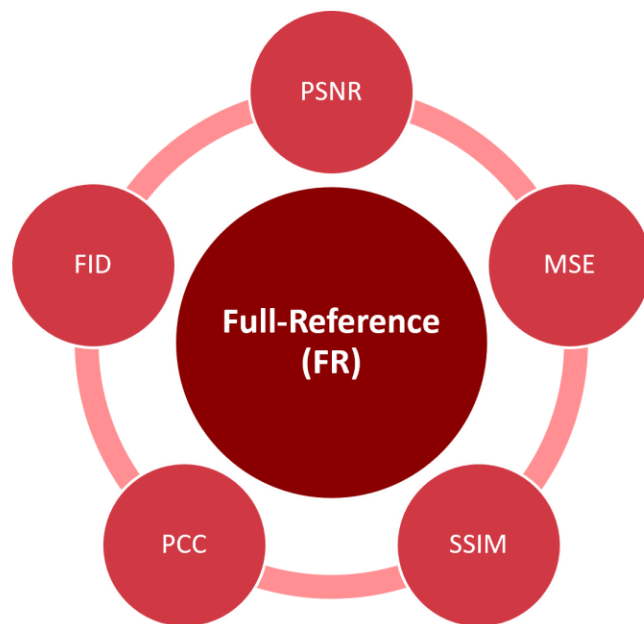
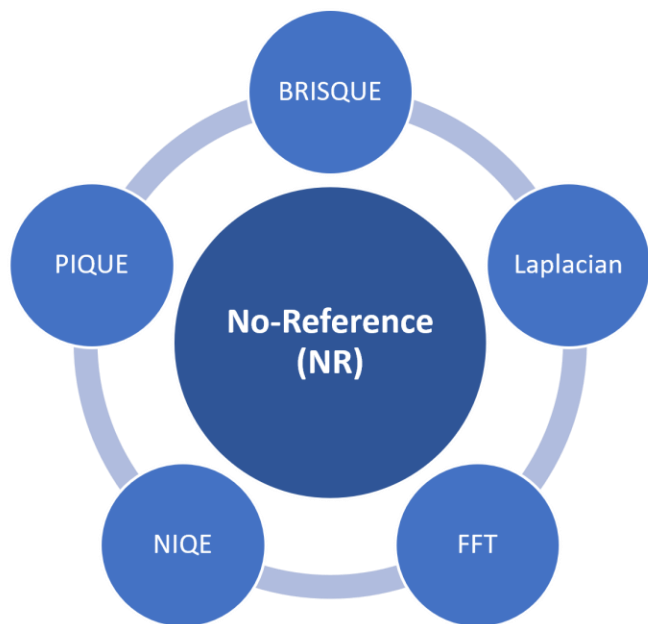
[Sara Concas, Simone Maurizio La Cava, Roberto Casula, Giulia Orrù, Giovanni Puglisi, Gian Luca Marcialis; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024]

Quality-based Artifact Modeling for Facial Deepfake Detection in Videos



[Sara Concas, Simone Maurizio La Cava, Roberto Casula, Giulia Orrù, Giovanni Puglisi, Gian Luca Marcialis; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)]

Quality-based Artifact Modeling for Facial Deepfake Detection in Videos



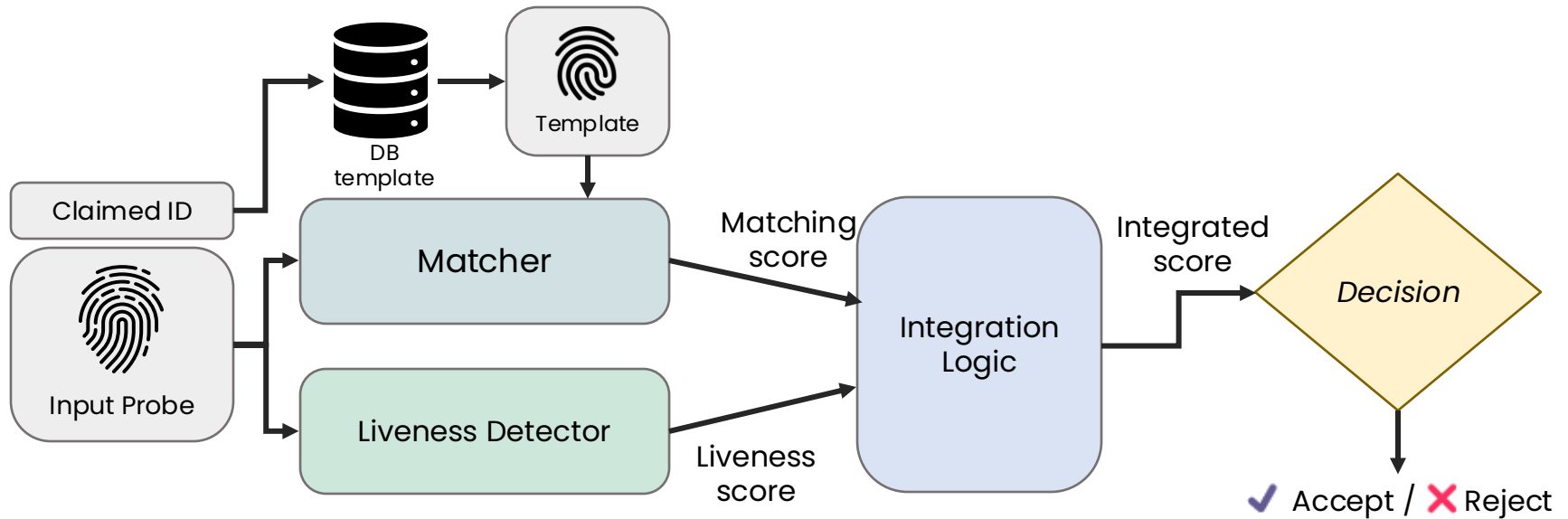
[Sara Concas, Simone Maurizio La Cava, Roberto Casula, Giulia Orrù, Giovanni Puglisi, Gian Luca Marcialis; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024]

Performance Comparison

- Intra-dataset
- Cross-dataset

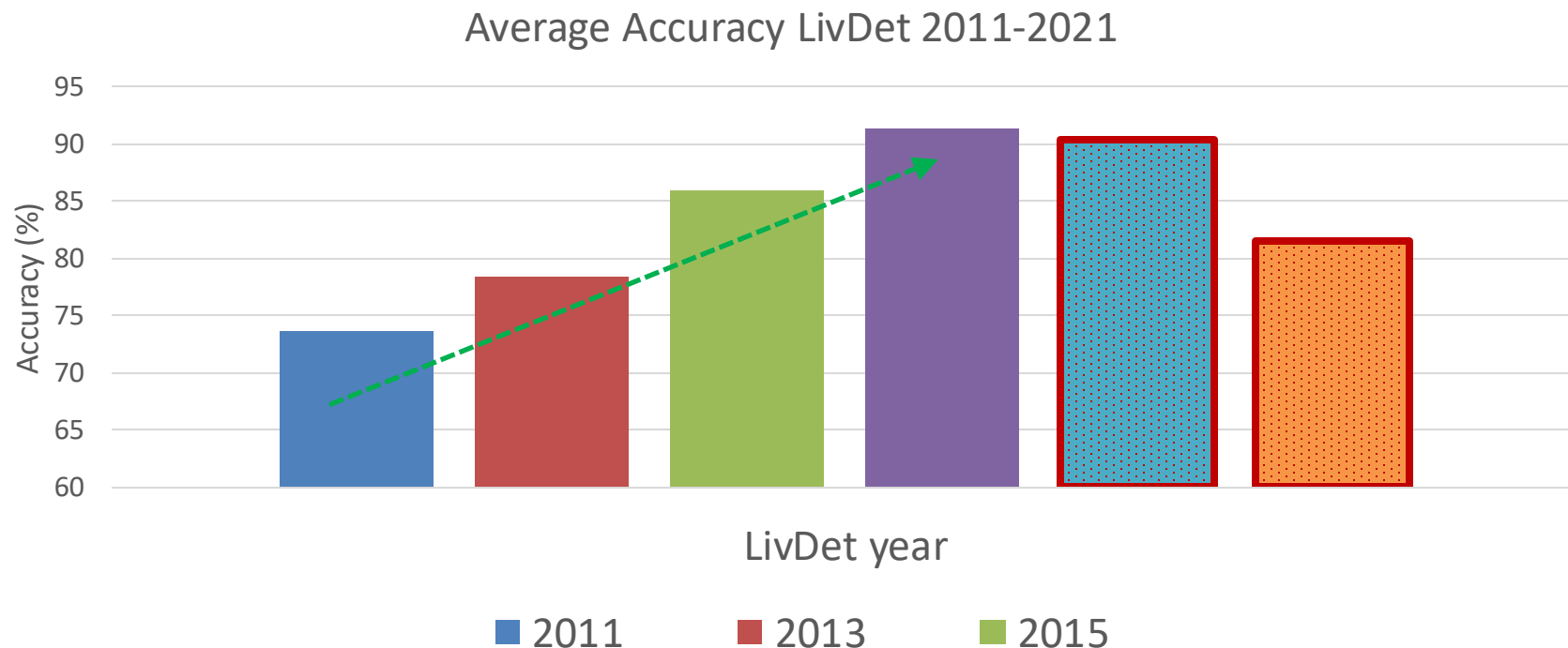
Method	Train Set	Test Set	Accuracy (%)	AUC (%)
V. L. L. Thing [1]	FF++	FF++	88.74	99.95
		DFD	84.2	94.4
		DFDC	68.9	76.3
	DFD	FF++	59.5	67.4
		DFD	99.5	100.00
		DFDC	71.1	78.5
	DFDC	FF++	59.3	74.2
		DFD	96.2	99.8
		DFDC	92.0	97.6
Yan et al. [2]	FF++	DFDC	-	77.0
Choi et al. [3]	FF++	DFD	-	96.1
Concas et al. (sAlfer Lab) Score-Level Fusion [8]	FF++	FF++	-	98.4
		DFDC	-	71.1
Gao et al. (sAlfer Lab) [9]	FF++	FF++	95.4	-
		DFDC	57.1	-
Concas et al. (sAlfer Lab) [10]	FF++	FF++	99.5	100.0

Biometrics: Integrated system scheme

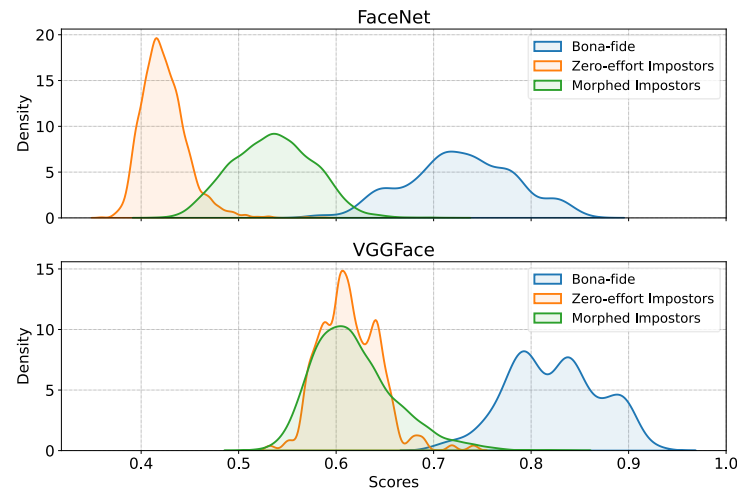
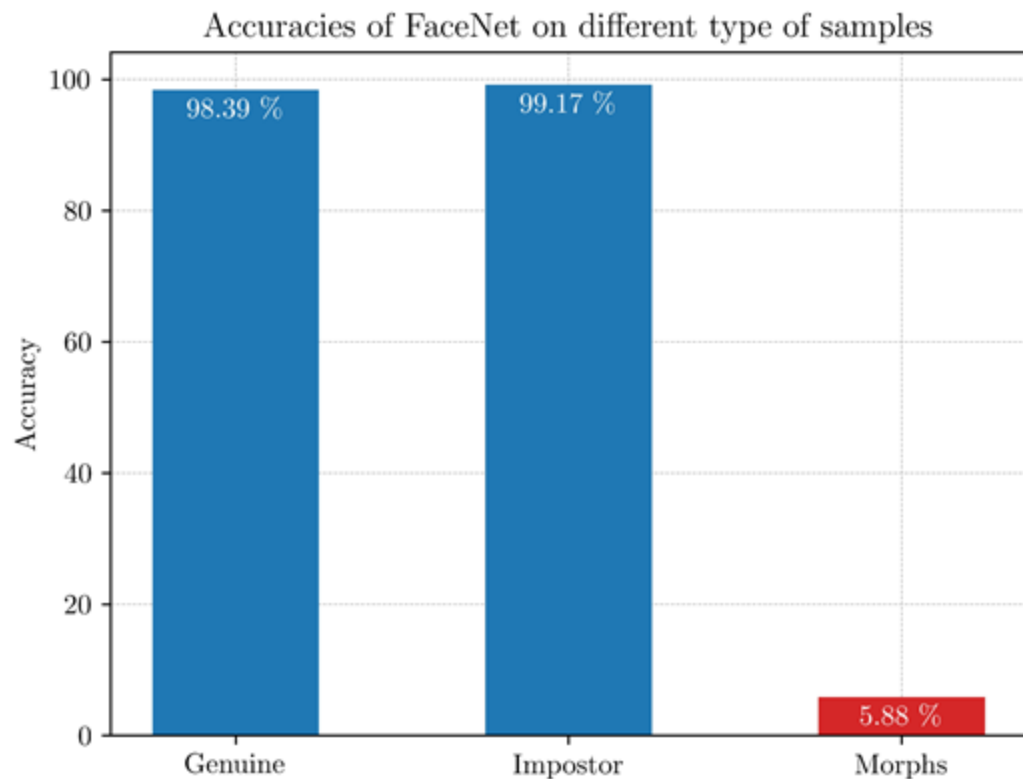


? What happens when we put detection and verification system together?

Integration: fingerprint presentation attack detection and matcher

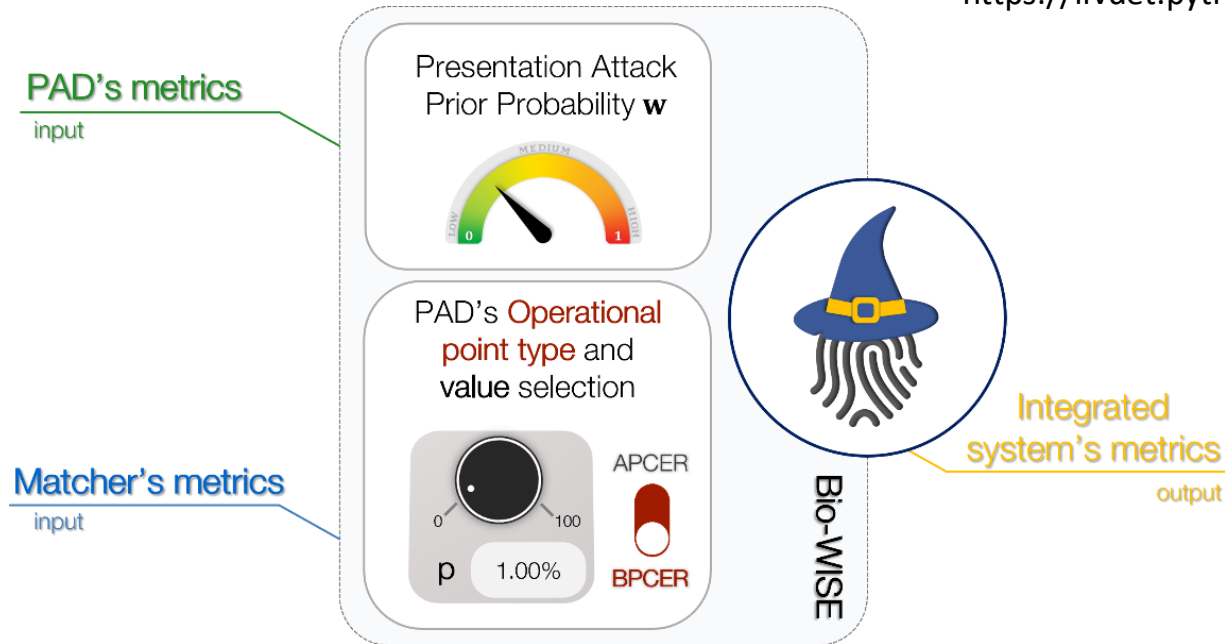


Integration: Morph as a serious threat to Face Recognition



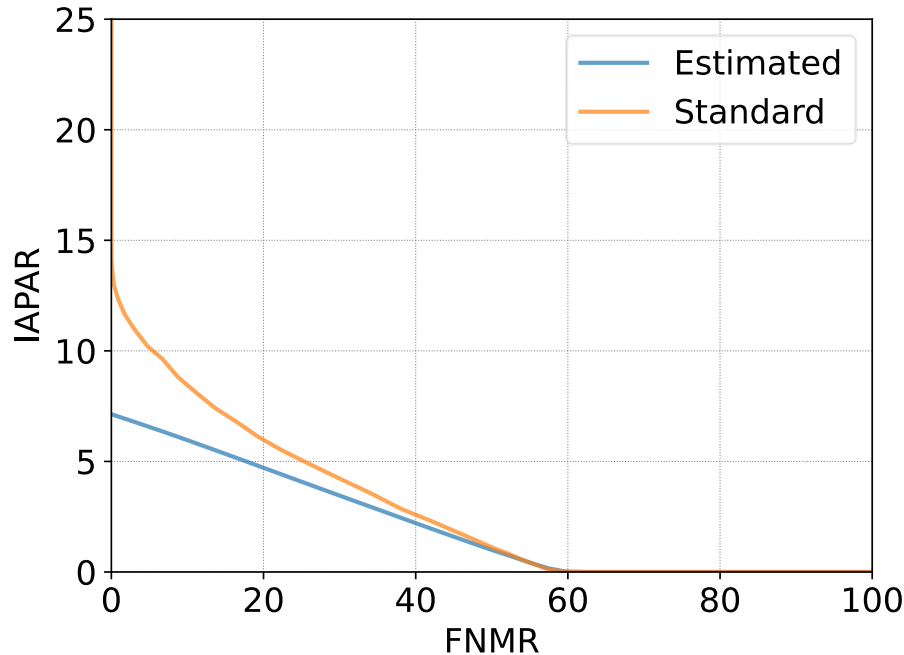
Bio-WISE: Integration Simulator

<https://livdet.pythonanywhere.com/>



Micheletto, M., Marcialis, G. L., Orrù, G., & Roli, F. (2021). Fingerprint recognition with embedded presentation attacks detection: are we ready?. *IEEE Transactions on Information Forensics and Security*, 16, 5338-5351.

BioWise (applied to MAD): Is The Simulator Reliable?



- A comparative study has been done, in order to assess the reliability of the proposed simulator
- It is possible to observe that the simulation tends to underestimate the value of the IAPAR, with a more marked difference for lower values of FNMR
- However, this error tends to disappear as the FNMR increases

Conclusions

**Generalization remains
the key challenge**

across all domains
(fingerprint, face
deepfake, audio,
morph), PAD and
deepfake detectors
trained on known
attacks struggle with
novel, unseen ones

**Dataset variability
amplifies the problem**

differences in
compression, resolution,
acquisition device and
generation model make
cross-dataset evaluation
harder than intra-
dataset

**Novel representations
improve robustness**

tensor-based,
inner/outer face
inconsistency,
texture/artifact
decomposition and
quality-based methods
show promising
generalization gains

**Score-level fusion
boosts performance**

combining multiple
detectors through
weighted or
classification-based
fusion consistently
outperforms single-
modal approaches

Open Challenges & Future Directions

Challenges

- Deepfake generation is rapidly evolving – detectors must keep pace with new architectures (diffusion models, etc.)
- Lack of large, diverse, and labeled real-world datasets for training and evaluation
- Integration with biometric systems adds complexity: PAD accuracy alone is insufficient
- Morph attacks remain a critical threat to border control and ID document systems

Future Directions

- Few-shot and zero-shot learning for rapid adaptation to novel attack types without full retraining
- Multi-modal fusion (face, audio, physiological signals) for more robust in-the-wild detection
- Explainability and certification of PAD systems for deployment in regulated environments
- Reliable simulation tools (Bio-WISE) to benchmark integrated biometric + PAD system performance



sAifer Lab

Joint lab on Safety and Security of AI

Thank you for your attention

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